

Lab 4: VibeCoding — Chat History

Brick Breaker (Repo 49_brick-breaker)

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LLM used: Claude (claude.ai) • Date: 01-10-2026

1. Fixing the code and adding all features

Dhruv (Prompt)

help me complete this lab. this is my github repo to work with:

[Attached: Lab_4_VibeCoding_Student_handout.pdf]

[Pasted the GitHub page of SETAPESU26/49_brick-breaker. Page navigation text omitted; README content below.]

Brick Breaker Game — a Brick Breaker (Breakout-style) clone using Pygame, provided partially working: paddle, ball, bricks, lives and score.

Task 1: Refine Collision Detection — the ball always bounces vertically off the paddle and bricks no matter which side it hit. Hitting a brick from the left or right should send the ball sideways. Make the bounce match the side actually hit.

Task 2: Implement Game Over Condition — show a screen with win/lose and the final score, then wait for input instead of printing to the console.

Task 3: Add Replay Option — after the end screen, let the user play again choosing Easy, Medium or Hard (ball speed/paddle size), or exit.

Task 4: Add Sound Feedback — sounds for brick breaking, ball hitting paddle or wall, and game-over/win.

Submission checklist: before video, after video, chat/LLM page link with complete history.

Claude (Response)

Claude cloned the repository, read every source file (main.py, game_engine.py, paddle.py, ball.py, brick.py), then made one commit per task:

```
3b89a2f Task 4: Add sound effects for bricks, paddle, walls, win and game over
c05c757 Task 3: Replay option with Easy/Medium/Hard difficulty selection
de95b40 Task 2: Add win/lose end screen with final score
ae0ab6c Task 1: Side-aware collision detection for paddle, bricks and walls
a87bb13 Initial Brick Breaker starter
```

All four tasks are done and tested (ran the game headless: collisions, the lose screen, and the difficulty restart all work).

What was changed

- Task 1 (collision):** A new `_collision_side()` function finds which side of the paddle or brick the ball hit by checking where the overlap is smallest. A hit on a top or bottom face flips `vy`; a hit on a left or right face flips `vx`. The ball is pushed out of the object so it can't clip into a neighbouring brick. The paddle bounce angle depends on where the ball lands on the paddle. Wall bounces use `abs()` so the ball can't get stuck in a wall.
- Task 2 (end screen):** The console `print` is replaced with a dark overlay showing "YOU WIN!" or "GAME OVER" and the final score, then waits for input.
- Task 3 (replay):** On the end screen, press 1/2/3 for Easy, Medium or Hard, or ESC to exit. Easy: slower ball, wider paddle. Hard: faster ball, narrower paddle. Current difficulty shown at the top.

- **Task 4 (sound):** New `game/sounds.py` generates beeps in code, so no audio files are needed. Different sounds for brick, paddle, walls, a win jingle and a lose tune. Plays silently if there's no audio device.

Steps: record the before video on the original code; run the fixed version and record the after video; push to your own repo (not SETAPESU26); export the chat history into Lab-4/.

2. Pushing to my GitHub repo

Dhruv (Prompt)

https://github.com/Dhruvjagadeesh/49_brick-breaker.git — give me the cd command

Claude (Response)

After unzipping, go into the folder and push:

```
cd 49_brick-breaker
git remote set-url origin https://github.com/Dhruvjagadeesh/49_brick-breaker.git
git push -u origin HEAD:main
```

If the push is rejected because the repo already has commits, use `--force`; safe here since the commits build on the same starter commit.

Dhruv (Prompt)

```
C:\Users\user>cd cd 49_brick-breaker
The system cannot find the path specified.
```

Claude (Response)

`cd` was typed twice, and the folder is probably in Downloads:

```
cd Downloads\49_brick-breaker
```

Dhruv (Prompt)

```
Game over! Final score: 6
...
C:\Users\user\49_brick-breaker>git push -u origin HEAD:main
Everything up-to-date
```

Claude (Response)

That folder is the **original** code — "Game over! Final score: 6" printed in the console is the old bug, and nothing new was pushed. That run works as the **before video**. Extract the fixed zip and check for the four Task commits with `git log --oneline`.

Dhruv (Prompt)

[Screenshot of Downloads folder showing game_engine, sounds and a 49_brick-breaker folder] right now its like this

Claude (Response)

```
cd C:\Users\user\Downloads\49_brick-breaker
git log --oneline
```

Dhruv (Prompt)

```
git log --oneline
```

```
de032ce (HEAD -> main) first commit
python main.py
python: can't open file '...\main.py': [Errno 2] No such file or directory
```

Claude (Response)

That folder is a different, empty repo. Run `dir` and check whether the zip was downloaded.

Dhruv (Prompt)

[dir output: the folder contains another 49_brick-breaker folder; 49_brick-breaker.zip is in Downloads]

Claude (Response)

The fixed version is in the nested folder:

```
cd 49_brick-breaker
git log --oneline
```

Dhruv (Prompt)

```
3b89a2f (HEAD -> main) Task 4: Add sound effects ...
c05c757 Task 3: Replay option ...
de95b40 Task 2: Add win/lose end screen ...
ae0ab6c Task 1: Side-aware collision detection ...
a87bb13 (origin/main) Initial Brick Breaker starter
```

Claude (Response)

Right place. Run `python main.py` and record the after video (side bounce, lose to show end screen, press 2 to replay), then push:

```
git remote set-url origin https://github.com/Dhruvjagadeesh/49_brick-breaker.git
git push -u origin HEAD:main --force
```

3. Adding the deliverables

Dhruv (Prompt)

[Screenshot: before.mp4 and after.mp4 in D:\projects\SE\LAB 4] its here rn

Claude (Response)

```
mkdir Lab-4
copy "D:\projects\SE\LAB 4\before.mp4" Lab-4\
copy "D:\projects\SE\LAB 4\after.mp4" Lab-4\
git add Lab-4
git commit -m "Add Lab-4 deliverables: before/after videos"
git push -u origin HEAD:main --force
```

Quotes are needed because **LAB 4** has a space. For the chat PDF, use Ctrl+P → Save as PDF.

Dhruv (Prompt)

[Screenshot: printed PDF came out blank] this is what came

Claude (Response)

Ctrl+P doesn't capture the chat because the page scrolls inside a container. Two options: (1) use the chat's **Share** link, which the README checklist asks for; (2) copy the chat into Word and save as PDF.

Dhruv (Prompt)

[Uploaded chat_history.docx] this is the doc rn, can u format it neatly

Claude (Response)

The pasted doc was missing most of the middle of the chat, so the complete transcript was rebuilt into this formatted document (PDF and Word), with the GitHub navigation clutter trimmed.

Dhruv (Prompt)

[Uploaded chat_history.docx again] pls format this doc neatly, it may have some missing parts fill it

Claude (Response)

Same file as before; the missing parts are filled in this document, and this final exchange was added so the transcript is complete.